

Yugeng ZHOU

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SUMMARY

Master's graduate in Scientific Computing at TU Berlin with strong foundations in Python, algorithmic problem solving, and building technical pipelines for experimentation and decision support. Hands-on experience across optimization workflows, performance evaluation, and LLM-assisted pipeline design with a focus on reliability, verification, and practical implementation. Interested in technical data operations roles where Python, data workflows, automation, and structured problem-solving are used to improve real-world AI and robotics systems.

SKILLS

- Programming: Python, C++, SQL
- Data & Workflow: data analysis, workflow automation, experiment design, quality checks, benchmarking
- AI & Tools: LLM-assisted workflows, output evaluation, verification, prompt-based automation
- Software & Systems: Git, Linux, modular pipeline design, reliability-focused development
- Optimization: Gurobi, SCIP, PySCIPOpt, pandas, NumPy

EXPERIENCE

AgentOpt / LLM-Assisted Optimization Pipeline

Independent Project, Feb. 2026 – Present

- Built an LLM-assisted pipeline to identify structured patterns in optimization models and generate task-specific code components.
- Combined structured prompting, domain knowledge, and iterative verification to improve output reliability and reduce unusable generations.
- Designed evaluation workflows to assess generation quality, correctness, and downstream usefulness in domain-specific problem settings.
- Explored how AI-generated components can be integrated into larger technical workflows for practical experimentation and analysis.

Cardinal Operations

Algorithm Intern, Mar. 2026 - Apr. 2026

- Built Python-based planning workflows for railway scheduling and resource assignment under complex operational constraints.
- Implemented batching and inheritance strategies to improve scalability and support more tractable decision workflows on large instances.
- Designed structured experiments to evaluate runtime, solution quality, and trade-offs across alternative modeling and heuristic choices.
- Translated operational requirements into implementable pipelines and decision-support logic.

Zuse Institute Berlin (ZIB)

Research Assistant Intern, Jan. 2024 - May. 2024

- Improved components of a parallel optimization solver and analyzed factors affecting runtime performance.
- Built visualizations and evaluation scripts to better understand search behavior and solver dynamics.
- Benchmarked heuristic methods on standard instances and supported performance-focused experimentation.

Zuse Institute Berlin (ZIB)

Research Assistant Intern, Nov. 2022 - Mar. 2023

- Built optimization models and data transformation workflows for transportation-related planning tasks.
- Supported preprocessing, benchmarking, and experimental analysis across different data and model formats.

EDUCATION

Technical University of Berlin, Berlin, Germany

M.Sc. in Scientific Computing, Nov 2020 – Oct 2025

- Relevant coursework: Combinatorial Optimization, Numerical Mathematics, Numerical Linear Algebra, Machine Learning
- Master's Thesis : *Branching on General Disjunctions using Distributed Memory Parallel Branch-and-Bound*
 - Implemented and evaluated a new decision strategy in a distributed-memory parallel search framework.

Harbin University of Science and Technology, Harbin, China

B.Sc. in Information and Computing Science, Sep 2015 – Jun 2019